

# INTRODUCTION TO UNIVERSITY MATHEMATICS

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ABSTRACT. Notes for the University of Helsinki course MAT11001, taught in Finnish from 31.08.2026–17.10.2026.

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## 1. COURSE INFORMATION

The official course information is available on the University of Helsinki course page.

## 2. LECTURE NOTES

2.1. **31.08.2026.** The first lecture was mostly spent going over the course structure and other general admin topics. We did go over some preliminary definitions and examples, which will follow.

2.1.1. *Notational conventions.*

**Definition 2.1.** An integer  $p > 1$  is *prime* if its only positive divisors are 1 and  $p$ .

**Definition 2.2.** The following basic sets of numbers will be used throughout:

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$\mathbb{Z}^+ = \mathbb{N} \setminus \{0\}$$

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$$

$\mathbb{R}$  = the set of real numbers

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}, \quad i^2 = -1$$

$$\mathbb{P} = \{p \in \mathbb{Z} : p \text{ is prime}\}$$

2.1.2. *Principle of mathematical induction.* Here we state the fundamental principle of mathematical induction without proof, and use it freely thereafter.

**Theorem 2.3** (Principle of mathematical induction). *Let  $n_0 \in \mathbb{N}$ , and let  $P(n)$  be a statement for each  $n \in \mathbb{N}$  with  $n \geq n_0$ . Suppose that*

- (1)  $P(n_0)$  is true, and
- (2) for every  $n \geq n_0$ ,  $P(n)$  implies  $P(n+1)$ .

*Then  $P(n)$  is true for every  $n \geq n_0$ .*

*Proof.* Omitted. □

**Example 2.4.** We prove by induction that

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

for every integer  $n \geq 1$ .

For  $n = 1$ , the identity holds because

$$\sum_{k=1}^1 k = 1 = \frac{1(1+1)}{2}.$$

Now suppose that the identity holds for some integer  $n \geq 1$ . Then

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

Thus the identity also holds for  $n+1$ , and hence for every integer  $n \geq 1$ . □

## 2.2. 03.09.2026.

2.2.1. *Divisibility of integers.* We start by defining the precise meaning of *even* and *odd* numbers.

**Definition 2.5.** A number  $m \in \mathbb{Z}$  is *even* if there exists  $k \in \mathbb{Z}$  such that

$$m = 2k.$$

On the other hand,  $m$  is *odd* if there exists a  $k \in \mathbb{Z}$  such that

$$m = 2k + 1.$$

**Definition 2.6** (Divisibility). A number  $a \in \mathbb{Z}$  is said to be *divisible* by a number  $b \in \mathbb{Z}, b \neq 0$ , if there exists a  $k \in \mathbb{Z}$  such that

$$a = kb.$$

In this case we also say that  $b$  *divides*  $a$ , and write

$$b \mid a.$$

If  $b$  does *not* divide  $a$ , we write

$$b \nmid a.$$

**Definition 2.7.** For  $n \in \mathbb{Z} \setminus \{0\}$ , we denote the set of integers that are divisible by  $n$  by

$$n\mathbb{Z} = \{z \in \mathbb{Z} : z = nk, \quad k \in \mathbb{Z}\}.$$

**2.2.2. Strong induction.** In some proofs it is useful to assume all preceding cases rather than only the immediately preceding case. This gives an equivalent form of the principle of induction, called *strong induction*.

**Theorem 2.8** (Strong induction). *Let  $n_0 \in \mathbb{N}$ , and let  $P(n)$  be a statement for each  $n \in \mathbb{N}$  with  $n \geq n_0$ . Suppose that*

- (1)  $P(n_0)$  is true, and
- (2) for every  $n \geq n_0$ , if  $P(k)$  is true for every  $k$  with  $n_0 \leq k \leq n$ , then  $P(n+1)$  is true.

*Then  $P(n)$  is true for every  $n \geq n_0$ .*

*Proof.* Omitted. □

As an example where strong induction is natural, let us prove that all natural numbers  $n \geq 2$  can be written as a product of *prime* numbers.

**Theorem 2.9.** *Every natural number  $n \geq 2$  can be written as a product of prime numbers.*

*Proof.* When  $n = 2$ , we have  $n \in \mathbb{P}$ , so the result holds trivially. Now suppose  $n \geq 2$  and the statement holds for all  $2 \leq k \leq n$ . If  $n+1 \in \mathbb{P}$ , the result holds trivially, so suppose  $n+1 \notin \mathbb{P}$ . Then we have

$$n+1 = ab,$$

for some  $2 \leq a, b \leq n$ . By the induction hypothesis,  $a$  and  $b$  can be written as products of primes, so

$$\begin{aligned} a &= p_1 p_2 \cdots p_i, & p_1, \dots, p_i &\in \mathbb{P}, \\ b &= q_1 q_2 \cdots q_j, & q_1, \dots, q_j &\in \mathbb{P}. \end{aligned}$$

Thus we have

$$n+1 = ab = p_1 p_2 \cdots p_i q_1 q_2 \cdots q_j.$$

The result follows from Theorem 2.8. □

**2.2.3. Sequences.**

**Definition 2.10.** A sequence of real numbers is a function  $z: \mathbb{N} \rightarrow \mathbb{R}$ , whose terms are denoted by  $z_n = z(n)$ . We write the sequence as

$$(z_n)_{n \in \mathbb{N}} = (z_0, z_1, z_2, \dots).$$

**Definition 2.11.** A sequence of real numbers may also be specified recursively. First, for some  $k \in \mathbb{Z}^+$ , we define the initial terms  $z_0, \dots, z_{k-1}$ . For each  $n \geq k$ , we then define  $z_n$  in terms of earlier terms  $z_0, \dots, z_{n-1}$ .

**2.3. 07.09.2026.** This lecture was mostly spent going through basic mathematical notation.

**2.3.1. Logical connectives.**

**Definition 2.12.** Let  $p$  and  $q$  be propositions. We define the following logical connectives:

- (1) The *conjunction*  $p \wedge q$ , read “ $p$  and  $q$ ”, is true exactly when both  $p$  and  $q$  are true.
- (2) The *disjunction*  $p \vee q$ , read “ $p$  or  $q$ ”, is true exactly when at least one of  $p$  and  $q$  is true.
- (3) The *implication*  $p \implies q$ , read “ $p$  implies  $q$ ”, is false exactly when  $p$  is true and  $q$  is false.
- (4) The *equivalence*  $p \iff q$ , read “ $p$  if and only if  $q$ ”, is true exactly when  $p$  and  $q$  have the same truth value.
- (5) The *negation*  $\neg p$ , read “not  $p$ ”, is true exactly when  $p$  is false.

**Definition 2.13.** Let  $p$  and  $q$  be propositions.

- (1) The *negated implication*  $p \not\Rightarrow q$ , read “ $p$  does not imply  $q$ ”, is true exactly when  $p$  is true and  $q$  is false.
- (2) The *negated equivalence*  $p \not\Leftrightarrow q$ , read “ $p$  is not equivalent to  $q$ ”, is true exactly when  $p$  and  $q$  have different truth values.

**Definition 2.14.** Let  $X$  be a set. A *predicate*  $P(x)$  on  $X$  is a statement containing a variable  $x$  whose truth value depends on the chosen value of  $x \in X$ . For each  $a \in X$ , substituting  $a$  for  $x$  gives the proposition  $P(a)$ , which is either true or false. For example,  $P(x) := x > 0$  is a predicate on  $\mathbb{R}$ ;  $P(2)$  is true, whereas  $P(-1)$  is false.

**Definition 2.15.** Let  $P(x)$  be a predicate on a set  $X$ .

- (1) The *universal quantification*  $\forall x \in X: P(x)$ , read “for every  $x \in X$ ,  $P(x)$ ”, is true exactly when  $P(a)$  is true for every  $a \in X$ .
- (2) The *existential quantification*  $\exists x \in X: P(x)$ , read “there exists an  $x \in X$  such that  $P(x)$ ”, is true exactly when  $P(a)$  is true for at least one  $a \in X$ .

**Definition 2.16.** The *negated existential quantification*  $\nexists x \in X: P(x)$ , read “there does not exist an  $x \in X$  such that  $P(x)$ ”, is true exactly when  $P(a)$  is false for every  $a \in X$ . Equivalently,

$$\nexists x \in X: P(x) \iff \forall x \in X: \neg P(x).$$

### 2.3.2. Basic set theory.

**Definition 2.17.** A *set* is a collection of distinct objects, called its *elements*. If an object  $x$  is an element of a set  $A$ , we write  $x \in A$ ; otherwise, we write  $x \notin A$ . The order in which the elements are listed, and any repetition among them, do not affect the set.

**Definition 2.18.** Two sets  $A$  and  $B$  are *equal*, written  $A = B$ , if they have exactly the same elements. Equivalently,

$$A = B \iff \forall x: (x \in A \iff x \in B).$$

**Definition 2.19.** The *empty set* is the set containing no elements. It is denoted by  $\emptyset$ .

**Definition 2.20.** Let  $A$  and  $B$  be sets.  $A$  and  $B$  are said to be *disjoint*, if  $A \cap B = \emptyset$ .

**Definition 2.21.** A *singleton* set is a set consisting of exactly **one** element.

**Definition 2.22.** Let  $P(x)$  be a predicate on a set  $X$ . The *set-builder notation*

$$A := \{x \in X: P(x)\}$$

defines  $A$  to be the set of precisely those elements  $x \in X$  for which  $P(x)$  is true. Equivalently, for every  $a \in X$ ,

$$a \in A \iff P(a).$$

**Definition 2.23.** A set  $A$  is a *subset* of a set  $B$  if every element of  $A$  is also an element of  $B$ . We then write  $A \subseteq B$ . If  $A$  is not a subset of  $B$ , we write  $A \not\subseteq B$ ; this means that some element of  $A$  is not an element of  $B$ . If  $A \subseteq B$  and  $A \neq B$ , then  $A$  is a *proper subset* of  $B$ , written  $A \subset B$ . Two sets are equal exactly when both containments hold:

$$A = B \iff A \subseteq B \wedge B \subseteq A.$$

**Definition 2.24.** Let  $A$  and  $B$  be subsets of a set  $X$ .

- (1) The *union* of  $A$  and  $B$  is

$$A \cup B := \{x \in X: x \in A \vee x \in B\}.$$

- (2) The *intersection* of  $A$  and  $B$  is

$$A \cap B := \{x \in X: x \in A \wedge x \in B\}.$$

(3) The *difference* of  $A$  and  $B$  is

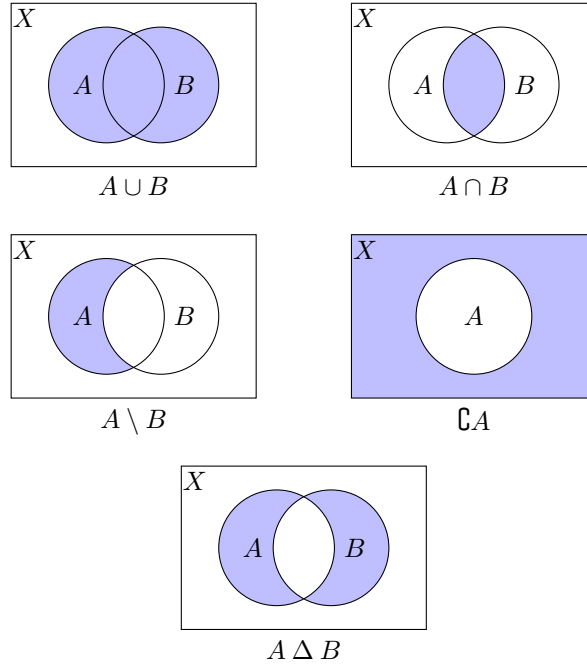
$$A \setminus B := \{x \in X : x \in A \wedge x \notin B\}.$$

(4) The *complement* of  $A$  in  $X$  is

$$\complement A := X \setminus A = \{x \in X : x \notin A\}.$$

(5) The *symmetric difference* of  $A$  and  $B$  is

$$A \Delta B := (A \setminus B) \cup (B \setminus A).$$



## 2.4. 10.09.2026.

### 2.4.1. Power sets.

**Definition 2.25.** The *power set* of a set  $A$ , denoted by  $\mathcal{P}(A)$ , is the set of all subsets of  $A$ :

$$\mathcal{P}(A) := \{B : B \subseteq A\}.$$

In particular,  $\emptyset \in \mathcal{P}(A)$  and  $A \in \mathcal{P}(A)$ .

**Definition 2.26.** Let  $A$  and  $B$  be sets. Their *Cartesian product* is the set of all ordered pairs whose first component belongs to  $A$  and whose second component belongs to  $B$ :

$$A \times B := \{(a, b) : a \in A \wedge b \in B\}.$$

More generally, for sets  $A_1, \dots, A_n$ , their Cartesian product is

$$A_1 \times \dots \times A_n := \{(a_1, \dots, a_n) : a_i \in A_i \text{ for every } i \in \{1, \dots, n\}\}.$$

If all the factors are the same set  $A$ , one commonly writes

$$A^n := \underbrace{A \times \dots \times A}_{n \text{ factors}}.$$

**Definition 2.27.** Let  $I$  be a countably infinite *index set*. A family of subsets of a set  $X$ , indexed by  $I$ , is denoted by  $(A_i)_{i \in I}$ . Its *union* and *intersection* are defined by

$$\bigcup_{i \in I} A_i := \{x \in X : \exists i \in I, x \in A_i\},$$

$$\bigcap_{i \in I} A_i := \{x \in X : \forall i \in I, x \in A_i\}.$$

Thus, an element belongs to the union if it belongs to at least one of the sets, and it belongs to the intersection if it belongs to every one of them.

If the family itself is denoted by  $\mathcal{A} = \{A_i : i \in I\}$ , one also writes  $\bigcup \mathcal{A}$  and  $\bigcap \mathcal{A}$ .

A common index set is  $I = \mathbb{N}$ . Since  $0 \in \mathbb{N}$  by our convention, one writes

$$\bigcup_{n=0}^{\infty} A_n = \bigcup_{n \in \mathbb{N}} A_n, \quad \bigcap_{n=0}^{\infty} A_n = \bigcap_{n \in \mathbb{N}} A_n.$$

For a family indexed by the positive integers, the corresponding notation is  $\bigcup_{n=1}^{\infty} A_n$  and  $\bigcap_{n=1}^{\infty} A_n$ . When the index range is clear from context, these are also abbreviated as  $\bigcup_n A_n$  and  $\bigcap_n A_n$ .

**2.4.2. Laws of set algebra.** Let  $A$ ,  $B$ , and  $C$  be subsets of a universal set  $X$ . The union and intersection operations are *commutative*:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A.$$

They are also *associative*:

$$(A \cup B) \cup C = A \cup (B \cup C), \quad (A \cap B) \cap C = A \cap (B \cap C).$$

The *distributive laws* are

$$\begin{aligned} A \cap (B \cup C) &= (A \cap B) \cup (A \cap C), \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C). \end{aligned}$$

The *De Morgan laws* relate complements to unions and intersections:

$$\begin{aligned} \mathbb{C}(A \cup B) &= \mathbb{C}A \cap \mathbb{C}B, \\ \mathbb{C}(A \cap B) &= \mathbb{C}A \cup \mathbb{C}B. \end{aligned}$$

Finally, set difference can be expressed using intersection and complement:

$$A \setminus B = A \cap \mathbb{C}B.$$

Equivalently, replacing  $B$  by  $C$  gives

$$A \setminus C = A \cap \mathbb{C}C.$$

**Theorem 2.28.** *Let  $A$  and  $B$  be subsets of a universal set  $X$ . Then*

$$A \subseteq B \iff \mathbb{C}B \subseteq \mathbb{C}A.$$

*Proof.* Suppose first that  $A \subseteq B$ , and let  $x \in \mathbb{C}B$ . Then  $x \notin B$ . Since every element of  $A$  belongs to  $B$ , it follows that  $x \notin A$ , and hence  $x \in \mathbb{C}A$ . Therefore  $\mathbb{C}B \subseteq \mathbb{C}A$ .

Conversely, suppose that  $\mathbb{C}B \subseteq \mathbb{C}A$ , and let  $x \in A$ . If  $x \notin B$ , then  $x \in \mathbb{C}B$ , so the assumed inclusion would imply that  $x \in \mathbb{C}A$ , contradicting  $x \in A$ . Thus  $x \in B$ , and therefore  $A \subseteq B$ .  $\square$

## 2.5. 14.09.2026.

**2.5.1. Proof by counterexample.** To disprove a universal claim, it is enough to find a single case in which the claim fails. Such a case is called a *counterexample*.

**Example 2.29.** Consider the claim that

$$(a + b)^2 = a^2 + b^2$$

for all real numbers  $a$  and  $b$ . Let  $a = b = 1$ . Then

$$(a + b)^2 = (1 + 1)^2 = 4, \quad a^2 + b^2 = 1^2 + 1^2 = 2.$$

Since  $4 \neq 2$ , this is a counterexample, and therefore the claim is false.

2.5.2. *Proof by contradiction.* In a proof by contradiction, we assume that the claim is false and show that this assumption leads to a contradiction.

**Example 2.30.** Suppose that  $A \subseteq B$  and  $A \not\subseteq C$ . We prove by contradiction that  $B \not\subseteq C$ .

Assume, to the contrary, that  $B \subseteq C$ . Since  $A \not\subseteq C$ , there exists an element  $x \in A$  such that  $x \notin C$ . But  $A \subseteq B$  implies that  $x \in B$ , and  $B \subseteq C$  then implies that  $x \in C$ . This contradicts  $x \notin C$ . Therefore  $B \not\subseteq C$ .

2.5.3. *Relations.*

**Definition 2.31.** A *relation* from a set  $A$  to a set  $B$  is a subset

$$R \subseteq A \times B.$$

If  $(a, b) \in R$ , we say that  $a$  is related to  $b$  and write  $aRb$ . A relation on  $A$  is a relation from  $A$  to itself, that is, a subset of  $A \times A$ .

**Definition 2.32.** Let  $R \subseteq A \times B$  be a relation. The *inverse relation* of  $R$  is

$$R^{-1} = \{(b, a) \in B \times A : (a, b) \in R\}.$$

Equivalently,  $bR^{-1}a$  if and only if  $aRb$ .

**Definition 2.33.** Let  $R$  be a relation on a set  $A$ . The relation  $R$  is

- (1) *reflexive* if  $xRx$  for every  $x \in A$ ;
- (2) *irreflexive* if  $(x, x) \notin R$  for every  $x \in A$ ;
- (3) *symmetric* if, for all  $x, y \in A$ ,  $xRy$  implies  $yRx$ ;
- (4) *antisymmetric* if, for all  $x, y \in A$ ,  $xRy$  and  $yRx$  imply  $x = y$ ;
- (5) *asymmetric* if, for all  $x, y \in A$ ,  $xRy$  implies  $(y, x) \notin R$ ;
- (6) *transitive* if, for all  $x, y, z \in A$ ,  $xRy$  and  $yRz$  imply  $xRz$ ; and
- (7) *total* if, for all  $x, y \in A$ , either  $xRy$  or  $yRx$ .

A relation is an *equivalence relation* if it is reflexive, symmetric, and transitive.

2.6. **17.09.2026.**

2.6.1. *Properties of inverse relations.*

**Theorem 2.34.** Let  $R, S \subseteq A \times B$  be relations. Then

- (a)  $(R^{-1})^{-1} = R$ ;
- (b)  $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$ ;
- (c)  $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$ ;
- (d)  $(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}$ .

*Proof.* For (a), let  $(a, b) \in A \times B$ . By the definition of the inverse relation,

$$(a, b) \in (R^{-1})^{-1} \iff (b, a) \in R^{-1} \iff (a, b) \in R.$$

For the remaining identities, let  $(b, a) \in B \times A$ . Then

$$\begin{aligned} (b, a) \in (R \cup S)^{-1} &\iff (a, b) \in R \cup S \\ &\iff (a, b) \in R \vee (a, b) \in S \\ &\iff (b, a) \in R^{-1} \vee (b, a) \in S^{-1} \\ &\iff (b, a) \in R^{-1} \cup S^{-1}, \end{aligned}$$

which proves (b). Similarly,

$$\begin{aligned} (b, a) \in (R \cap S)^{-1} &\iff (a, b) \in R \wedge (a, b) \in S \\ &\iff (b, a) \in R^{-1} \wedge (b, a) \in S^{-1} \\ &\iff (b, a) \in R^{-1} \cap S^{-1}, \end{aligned}$$

proving (c). Finally,

$$\begin{aligned} (b, a) \in (R \setminus S)^{-1} &\iff (a, b) \in R \wedge (a, b) \notin S \\ &\iff (b, a) \in R^{-1} \wedge (b, a) \notin S^{-1} \\ &\iff (b, a) \in R^{-1} \setminus S^{-1}, \end{aligned}$$

which proves (d). In each case, the two sets have exactly the same elements.  $\square$

*Remark 2.35.* In (d), the handwritten notes have an extra inverse on  $S$  inside the parentheses. The corrected identity is stated above. If the expression  $(R \setminus S^{-1})^{-1}$  is used literally, its value is  $R^{-1} \setminus S$ .

### 2.6.2. Composition of relations.

**Definition 2.36.** Let  $R \subseteq X \times Y$  and  $S \subseteq Y \times Z$ . Their *composite relation* is the relation from  $X$  to  $Z$  defined by

$$S \circ R := \{(x, z) \in X \times Z : \exists y \in Y, (x, y) \in R \wedge (y, z) \in S\}.$$

Equivalently,

$$x(S \circ R)z \iff \exists y \in Y : xRy \wedge ySz.$$

Thus  $S \circ R$  means that we first follow  $R$ , then  $S$ .

**Theorem 2.37** (Associativity of composition). *Let  $R \subseteq X \times Y$ ,  $S \subseteq Y \times Z$ , and  $T \subseteq Z \times U$ . Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

*Proof.* Let  $(x, u) \in X \times U$ . If  $(x, u) \in T \circ (S \circ R)$ , there is a  $z \in Z$  such that  $x(S \circ R)z$  and  $zTu$ . The first relation supplies a  $y \in Y$  with  $xRy$  and  $ySz$ . Since  $ySz$  and  $zTu$ , we have  $y(T \circ S)u$ , and hence  $(x, u) \in (T \circ S) \circ R$ .

Conversely, if  $(x, u) \in (T \circ S) \circ R$ , there is a  $y \in Y$  such that  $xRy$  and  $y(T \circ S)u$ . Choose  $z \in Z$  with  $ySz$  and  $zTu$ . Then  $x(S \circ R)z$ , so  $(x, u) \in T \circ (S \circ R)$ . Both inclusions hold, proving the equality.  $\square$

**2.6.3. Equivalence relations and equivalence classes.** Recall that a relation  $\sim$  on a set  $X$  is an *equivalence relation* if it is reflexive, symmetric, and transitive.

**Problem 2.38** (Lecture exercise). Let  $A = \{2, 3, 4, 5, 6\}$ , and define a relation  $R$  on  $A$  by

$$xRy \iff x \text{ and } y \text{ have a common prime factor.}$$

Is  $R$  an equivalence relation?

*Solution.* The relation is reflexive: each  $x \in A$  has a prime factor, and that prime divides both occurrences of  $x$ . More explicitly, 2 divides 2, 4, 6, 3 divides 3, and 5 divides 5.

It is also symmetric. If  $xRy$ , some prime  $p$  divides both  $x$  and  $y$ . The same prime divides both  $y$  and  $x$ , so  $yRx$ .

However, it is not transitive. We have  $4R6$ , since 2 divides both 4 and 6, and  $6R3$ , since 3 divides both 6 and 3. But 4 and 3 have no common prime factor: their only prime factors are 2 and 3, respectively. Thus  $(4, 3) \notin R$ . Consequently,  $R$  is not an equivalence relation.  $\square$

**Definition 2.39.** Let  $\sim$  be an equivalence relation on a set  $X$ , and let  $a \in X$ . The *equivalence class* of  $a$  is

$$[a]_{\sim} := \{x \in X : x \sim a\}.$$

The element  $a$  is called a *representative* of this class. The set of all equivalence classes is the *quotient set*

$$X/\sim := \{[a]_{\sim} : a \in X\}.$$

The following consequence explains why equivalence classes divide a set into separate parts; it supplements the definitions in the lecture.

**Proposition 2.40.** Let  $\sim$  be an equivalence relation on  $X$ . For every  $a, b \in X$ ,

- (a)  $a \in [a]_{\sim}$ ;
- (b)  $a \sim b$  if and only if  $[a]_{\sim} = [b]_{\sim}$ ;
- (c) the classes  $[a]_{\sim}$  and  $[b]_{\sim}$  are either equal or disjoint.

Therefore the equivalence classes form a partition of  $X$ : they are nonempty, their union is  $X$ , and distinct classes are disjoint.

*Proof.* Reflexivity gives  $a \sim a$ , proving (a).

For (b), suppose  $a \sim b$ . If  $x \in [a]_{\sim}$ , then  $x \sim a$ , so transitivity gives  $x \sim b$ . Hence  $[a]_{\sim} \subseteq [b]_{\sim}$ . Symmetry gives  $b \sim a$ , and the same argument gives the reverse inclusion. Conversely, if the classes are equal, then (a) gives  $a \in [a]_{\sim} = [b]_{\sim}$ , which means  $a \sim b$ .

For (c), if the classes intersect, choose an element  $c$  in their intersection. Then  $c \sim a$  and  $c \sim b$ . By symmetry,  $a \sim c$ , so transitivity gives  $a \sim b$ . Part (b) now gives equality of the classes. Thus distinct classes must be disjoint. Finally, every class is a subset of  $X$ , and (a) shows that each element of  $X$  belongs to a nonempty class. Their union is therefore all of  $X$ , proving the partition claim.  $\square$

#### 2.6.4. Functions.

**Definition 2.41.** Let  $X$  and  $Y$  be nonempty sets. A *function* (or *mapping*)  $f$  from  $X$  to  $Y$  assigns to each element  $x \in X$  exactly one element  $y \in Y$ . We write

$$f: X \longrightarrow Y, \quad y = f(x).$$

The set  $X$  is the *domain*, and  $Y$  is the *codomain*.

In terms of relations, the graph of  $f$  is

$$G_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y.$$

A relation  $F \subseteq X \times Y$  is the graph of a function precisely when, for every  $x \in X$ , there exists exactly one  $y \in Y$  such that  $(x, y) \in F$ . Existence ensures that the function is defined at every input, and uniqueness ensures that its value is unambiguous.

#### 2.7. 21.09.2026.

**2.7.1. Inverse functions, images, and preimages.** The inverse relation of a function  $f: X \rightarrow Y$  is a relation from  $Y$  to  $X$ , but it is not generally a function. If this inverse relation is a function from  $Y$  to  $X$ , it is called the *inverse function* of  $f$ , denoted by  $f^{-1}$ .

**Definition 2.42.** Let  $f: X \rightarrow Y$  be a function. The *image* of a subset  $A \subseteq X$  is

$$f(A) := \{f(x) : x \in A\}.$$

The *preimage* of a subset  $B \subseteq Y$  is

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

**Remark 2.43.** The preimage  $f^{-1}(B)$  is defined for every function  $f$ , even when  $f$  has no inverse function.

## 2.7.2. Images and preimages of unions and intersections.

**Theorem 2.44.** Let  $f: X \rightarrow Y$ . For all  $A, B \subseteq X$ ,

- (i)  $f(A \cup B) = f(A) \cup f(B)$ ;
- (ii)  $f(A \cap B) \subseteq f(A) \cap f(B)$ .

**Theorem 2.45.** Let  $f: X \rightarrow Y$ . For all  $A, B \subseteq Y$ ,

- (i)  $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$ ;
- (ii)  $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$ .

*Proof.* Let  $A, B \subseteq Y$  and  $x \in X$ . For (i),

$$\begin{aligned} x \in f^{-1}(A \cup B) &\iff f(x) \in A \vee f(x) \in B \\ &\iff x \in f^{-1}(A) \vee x \in f^{-1}(B) \\ &\iff x \in f^{-1}(A) \cup f^{-1}(B). \end{aligned}$$

Similarly, for (ii),

$$\begin{aligned} x \in f^{-1}(A \cap B) &\iff f(x) \in A \wedge f(x) \in B \\ &\iff x \in f^{-1}(A) \wedge x \in f^{-1}(B) \\ &\iff x \in f^{-1}(A) \cap f^{-1}(B). \end{aligned}$$

Thus both identities hold. □

## 2.7.3. Injective, surjective, and bijective functions.

**Definition 2.46.** A function  $f: X \rightarrow Y$  is *injective* if distinct inputs have distinct images: for all  $x_1, x_2 \in X$ ,

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

**Definition 2.47.** A function  $f: X \rightarrow Y$  is *surjective* if, for every  $y \in Y$ , there exists  $x \in X$  such that  $f(x) = y$ . Equivalently,  $f(X) = Y$ .

**Definition 2.48.** A function  $f: X \rightarrow Y$  is *bijective* if it is both injective and surjective.

**Theorem 2.49.** A function  $f: X \rightarrow Y$  has an inverse function  $f^{-1}: Y \rightarrow X$  if and only if  $f$  is bijective.

## 2.7.4. Lecture exercise: the floor function.

**Problem 2.50** (Lecture exercise). Let

$$f(x) = \lfloor x \rfloor = \max \{n \in \mathbb{Z} : n \leq x\}.$$

Determine whether  $f$  is injective, surjective, or bijective for each of the following choices of domain and codomain:

- (a)  $f: \mathbb{R} \rightarrow \mathbb{Z}$ ;
- (b)  $f: \mathbb{Z} \rightarrow \mathbb{R}$ ;
- (c)  $f: \mathbb{Z} \rightarrow \mathbb{Z}$ ;
- (d)  $f: \mathbb{R} \rightarrow \mathbb{R}$ .

*Solution.* (a) The function is not injective, since  $f(1) = f(1.5) = 1$ , although  $1 \neq 1.5$ . It is surjective: for every  $y \in \mathbb{Z}$ , any  $x \in [y, y+1)$  satisfies  $f(x) = y$ . Therefore it is not bijective.

- (b) For every  $x \in \mathbb{Z}$ , we have  $f(x) = x$ , so the function is injective. It is not surjective, since  $1/2 \in \mathbb{R}$  is not the image of any integer. Therefore it is not bijective.

- (c) Here  $f(x) = x$  for all  $x \in \mathbb{Z}$ , so  $f$  is the identity function on  $\mathbb{Z}$ . It is both injective and surjective, and hence bijective.
- (d) The function is neither injective nor surjective:  $f(1) = f(1.5)$ , and  $1/2$  is not in its image. Consequently, it is not bijective.  $\square$

## 2.8. 24.09.2026.

### 2.8.1. Images followed by preimages.

**Theorem 2.51.** Let  $f: X \rightarrow Y$ ,  $A \subseteq X$ , and  $B \subseteq Y$ . Then

- (i)  $A \subseteq f^{-1}(f(A))$ ;
- (ii)  $f(f^{-1}(B)) \subseteq B$ .

*Proof.* For (i), let  $x \in A$ . Then  $f(x) \in f(A)$ , so the definition of preimage gives  $x \in f^{-1}(f(A))$ .

For (ii), let  $y \in f(f^{-1}(B))$ . There exists  $x \in f^{-1}(B)$  with  $y = f(x)$ . Since  $x \in f^{-1}(B)$  means that  $f(x) \in B$ , we obtain  $y \in B$ .  $\square$

*Remark 2.52.* These inclusions need not be equalities. For  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x^2$ , and  $A = \{1\}$ , we have  $f^{-1}(f(A)) = \{-1, 1\}$ . For the same function and  $B = \{-1\}$ , we have  $f(f^{-1}(B)) = \emptyset$ . Here  $f^{-1}$  denotes a preimage, not an inverse function.

### 2.8.2. Composition of functions.

**Definition 2.53.** Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be functions. Their *composite function* is

$$g \circ f: X \rightarrow Z, \quad (g \circ f)(x) = g(f(x)) \quad (x \in X).$$

Thus we apply  $f$  first and  $g$  second.

**Theorem 2.54.** If  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  are bijections, then  $g \circ f: X \rightarrow Z$  is a bijection.

*Proof.* To prove injectivity, suppose that  $(g \circ f)(x_1) = (g \circ f)(x_2)$ . Then  $g(f(x_1)) = g(f(x_2))$ . Injectivity of  $g$  gives  $f(x_1) = f(x_2)$ , and injectivity of  $f$  gives  $x_1 = x_2$ .

To prove surjectivity, let  $z \in Z$ . Since  $g$  is surjective, there is  $y \in Y$  with  $g(y) = z$ . Since  $f$  is surjective, there is  $x \in X$  with  $f(x) = y$ . Thus  $(g \circ f)(x) = g(y) = z$ .  $\square$

### 2.8.3. Identity maps and inverse functions.

**Definition 2.55.** The *identity map* on a set  $X$  is the function

$$\text{id}_X: X \rightarrow X, \quad \text{id}_X(x) = x \quad (x \in X).$$

**Theorem 2.56.** Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow X$ . Then  $g$  is the inverse function of  $f$  if and only if

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y.$$

*Proof.* Suppose first that  $g = f^{-1}$ . For every  $x \in X$ , the inverse sends  $f(x)$  back to  $x$ , so  $g(f(x)) = x$ . For every  $y \in Y$ , the value  $g(y)$  is the unique element mapped to  $y$  by  $f$ , so  $f(g(y)) = y$ . These are the required identities.

Conversely, suppose both identities hold. If  $f(x_1) = f(x_2)$ , applying  $g$  gives

$$x_1 = g(f(x_1)) = g(f(x_2)) = x_2,$$

so  $f$  is injective. For every  $y \in Y$ , the element  $g(y) \in X$  satisfies  $f(g(y)) = y$ , so  $f$  is surjective. Thus each  $y$  has exactly one preimage, namely  $g(y)$ , proving that  $g = f^{-1}$ .  $\square$

**Corollary 2.57.** The inverse of a bijection is a bijection.

*Proof.* Let  $f: X \rightarrow Y$  be a bijection. If  $f^{-1}(y_1) = f^{-1}(y_2)$ , applying  $f$  gives  $y_1 = y_2$ , so  $f^{-1}$  is injective. For any  $x \in X$ , set  $y = f(x)$ . Then  $f^{-1}(y) = x$ , so  $f^{-1}$  is also surjective.  $\square$

#### 2.8.4. Infinite sets and their cardinalities.

**Definition 2.58.** Sets  $A$  and  $B$  have *equal cardinality* (or are *equinumerous*) if there exists a bijection  $A \rightarrow B$ . We write  $A \sim B$ .

**Definition 2.59.** A set  $A$  is *countably infinite* if  $A \sim \mathbb{N}$ . Equivalently, its elements can be listed as  $a_0, a_1, a_2, \dots$ , with every element appearing exactly once.

*Remark 2.60.* The Finnish term *numeroituva* in this lecture is defined by  $A \sim \mathbb{N}$ . We use “countably infinite” for this precise meaning; “countable” is also commonly used to include finite sets. Recall that  $\mathbb{N} = \{0, 1, 2, \dots\}$  in these notes.

For this discussion, we also allow functions with empty domain: the empty relation is the unique function  $\emptyset \rightarrow B$  for any set  $B$ . In particular,  $\text{id}_\emptyset$  is a bijection from  $\emptyset$  to itself. This extends the earlier convention of discussing functions between nonempty sets and allows us to include all subsets below.

**Theorem 2.61.** *Let  $X$  be a set. Equal cardinality is an equivalence relation on  $\mathcal{P}(X)$ .*

*Proof.* We verify reflexivity, symmetry, and transitivity.

*Reflexivity.* Let  $A \subseteq X$ . The identity map  $\text{id}_A: A \rightarrow A$  is injective, since equal outputs are equal inputs, and surjective, since each  $a \in A$  is its own image. Hence  $A \sim A$ , including when  $A = \emptyset$ .

*Symmetry.* Let  $A, B \subseteq X$  and suppose  $A \sim B$ . Choose a bijection  $f: A \rightarrow B$ . By corollary 2.57, its inverse  $f^{-1}: B \rightarrow A$  is a bijection, so  $B \sim A$ .

*Transitivity.* Let  $A, B, C \subseteq X$ , and suppose  $A \sim B$  and  $B \sim C$ . Choose bijections  $f: A \rightarrow B$  and  $g: B \rightarrow C$ . By theorem 2.54,  $g \circ f: A \rightarrow C$  is a bijection. Therefore  $A \sim C$ .

Thus  $\sim$  is an equivalence relation on  $\mathcal{P}(X)$ .  $\square$